

Day 12 BI Presentation

Linear Regression, Regularization, GLMs, Logistic Regression and advanced diagnostics

Linear and Logistic Regression

Ruturaj Mokashi

Executive storyline

From model mechanics to BI-ready recommendations

Notebook 1 proves why multiple regression is stronger than TV-only regression for Sales.

Notebook 2 shows how regularization controls unstable coefficients and supports feature selection.

Notebook 3 explains why count targets need Poisson or Negative Binomial thinking.

Notebook 4 turns Logistic Regression probabilities into campaign decisions using thresholds, calibration and lift.

Notebook 5 closes advanced diagnostics and explains how to trust or challenge model outputs.

Advertising

Multiple R² 0.8870

All channels together explain Sales much better than TV alone.

Regularization

LassoCV RMSE 52.9196

Regularization improves stability and removes weak coefficients.

Bike Sharing

NB deviance 4.213

Negative Binomial is more count-aware when overdispersion exists.

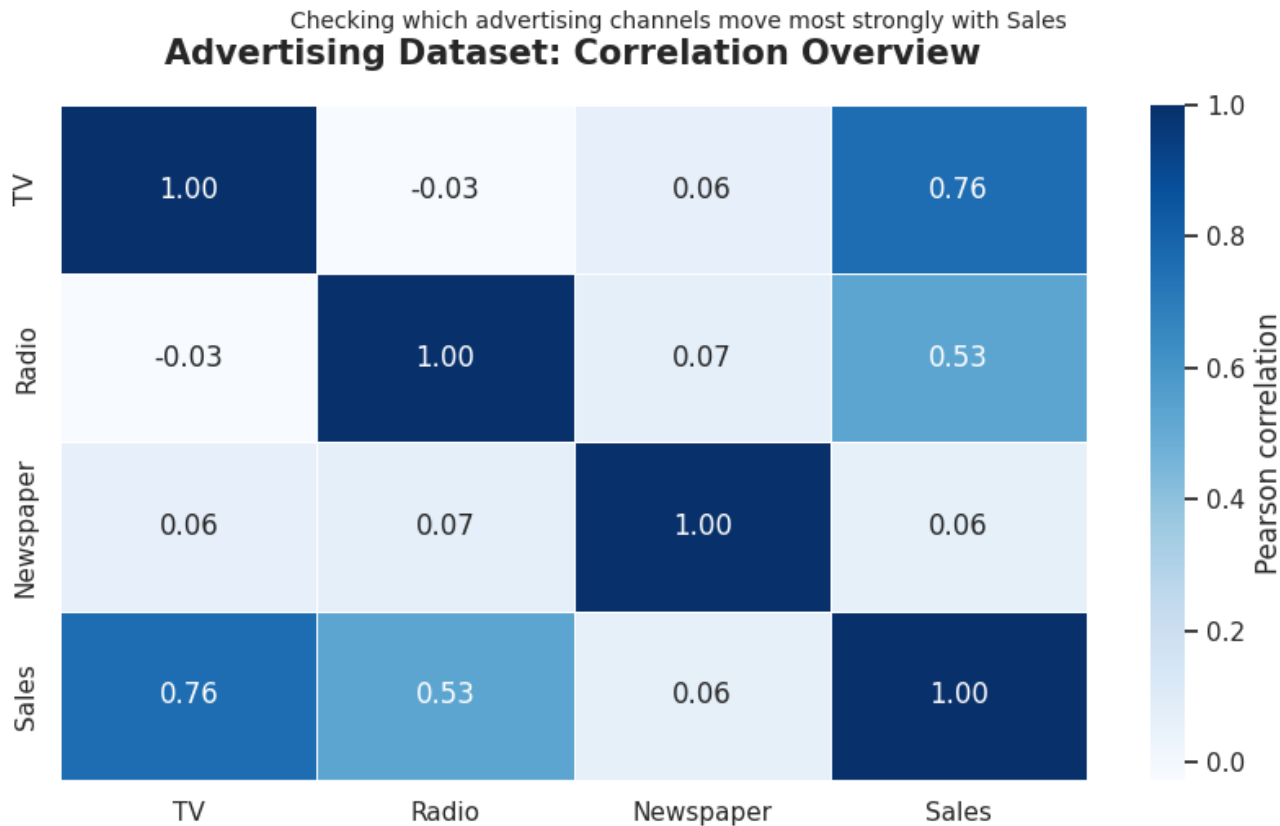
Campaign

ROC-AUC 0.614

Probability ranking is useful but threshold tuning is required.

11. BI-Ready Correlation Heatmap The heatmap makes the correl

Notebook chart evidence and business interpretation



How to read this chart

11. BI-Ready Correlation Heatmap The heatmap makes the correlation table easier to read visually. What to look for: darker or stronger cells show stronger relationships. For this assignment, we especially care about how TV, Radio, and Newspaper relate to Sales.

BI interpretation

Use this visual to connect model output with a decision: better fit, stronger driver, more reliable probabilities, a safer threshold, or a clearer diagnostic risk. The chart should support a recommendation rather than remain a technical output.

12. MVD Task 1 — TV Budget vs Sales with Regression Line

Notebook chart evidence and business interpretation

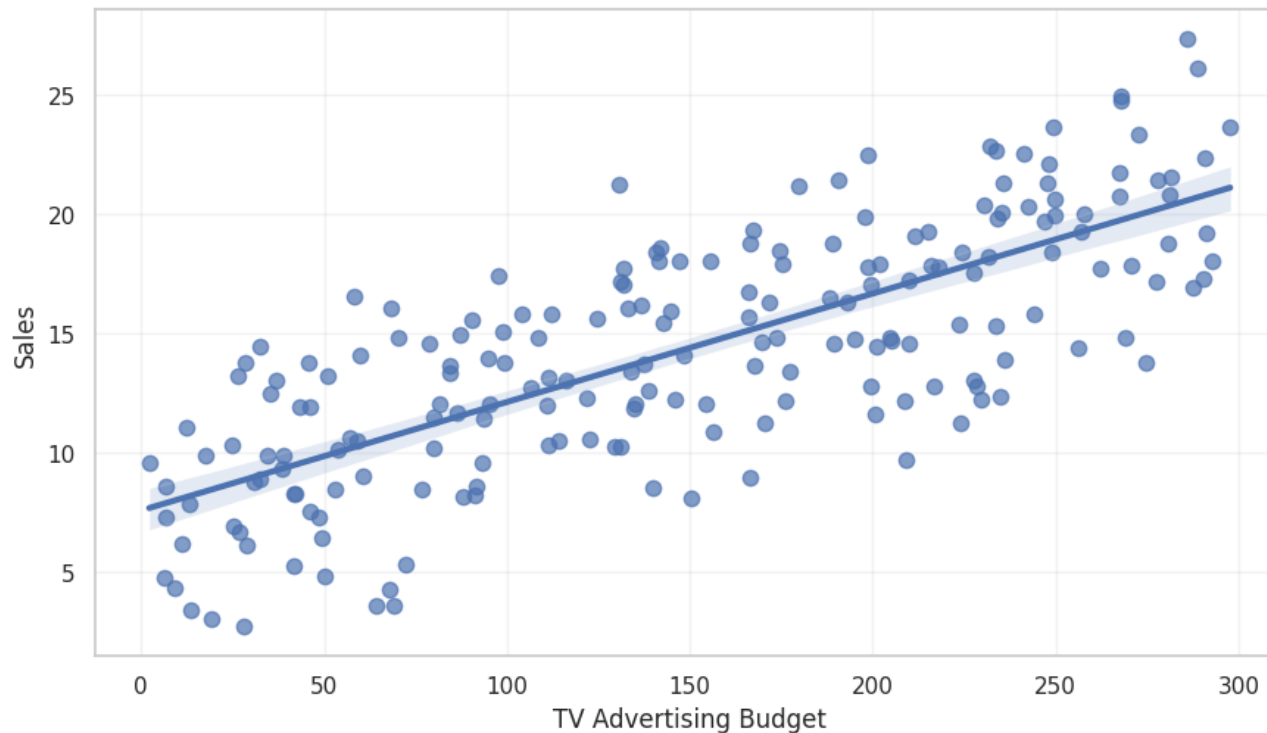
How to read this chart

12. MVD Task 1 — TV Budget vs Sales with Regression Line This is the first required MVD plot: TV budget vs Sales with a regression line. Why this plot matters: before building the model, we want to see whether a straight-line relationship is reasonable. If the points roughly rise upward, a linear model is a sensible first baseline. Required method: the MVD explicitly asks for `sns.regplot`, so we use it here.

BI interpretation

Use this visual to connect model output with a decision: better fit, stronger driver, more reliable probabilities, a safer threshold, or a clearer diagnostic risk. The chart should support a recommendation rather than remain a technical output.

Required MVD scatter plot with regression line using `seaborn.regplot`
TV Advertising Budget vs Sales



16. Actual vs Predicted Chart for Simple Model

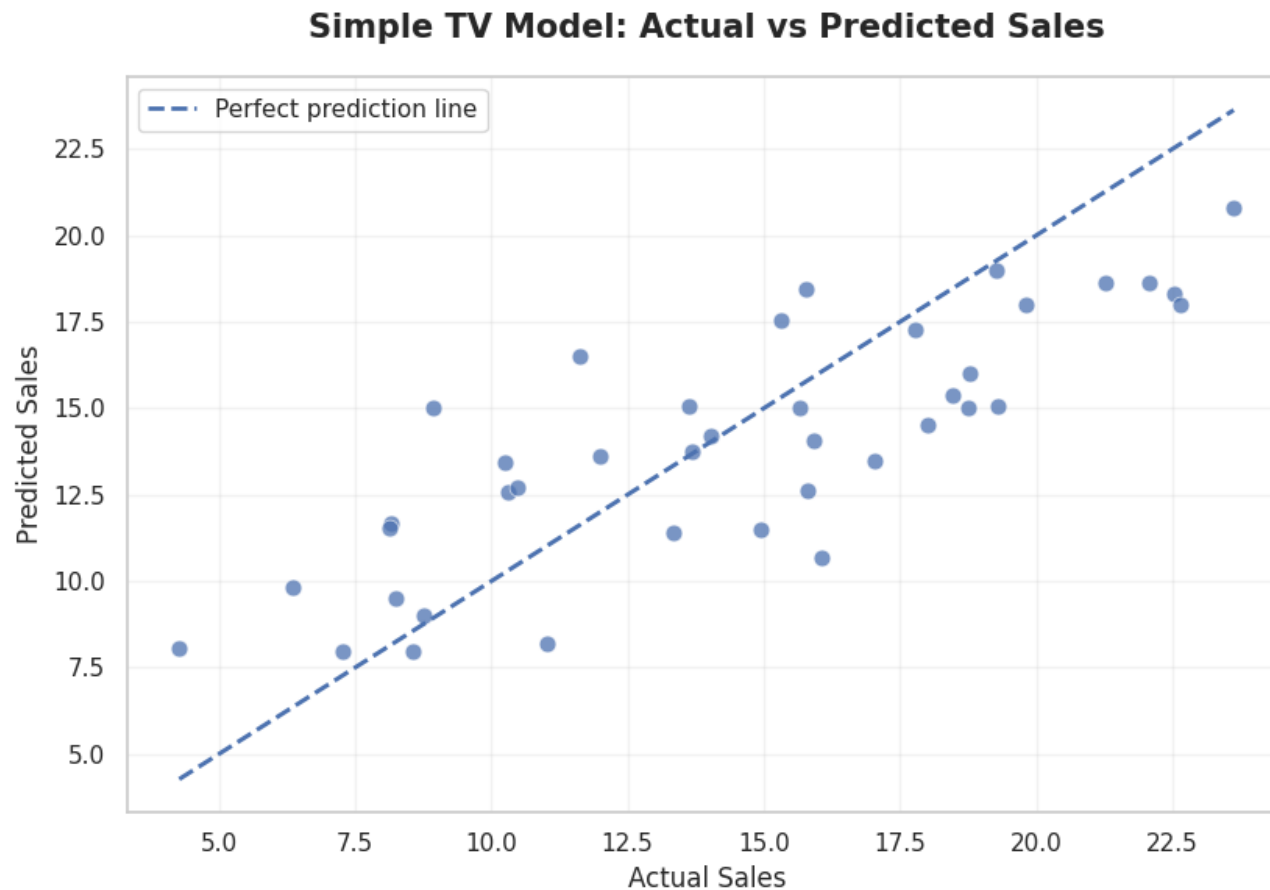
Notebook chart evidence and business interpretation

How to read this chart

16. Actual vs Predicted Chart for Simple Model This chart checks how close the TV-only predictions are to the real Sales values. How to read it: points closer to the diagonal line are better predictions. Points far away from the line are larger errors.

BI interpretation

Use this visual to connect model output with a decision: better fit, stronger driver, more reliable probabilities, a safer threshold, or a clearer diagnostic risk. The chart should support a recommendation rather than remain a technical output.



21. BI-Ready Model Comparison Chart

Notebook chart evidence and business interpretation

How to read this chart

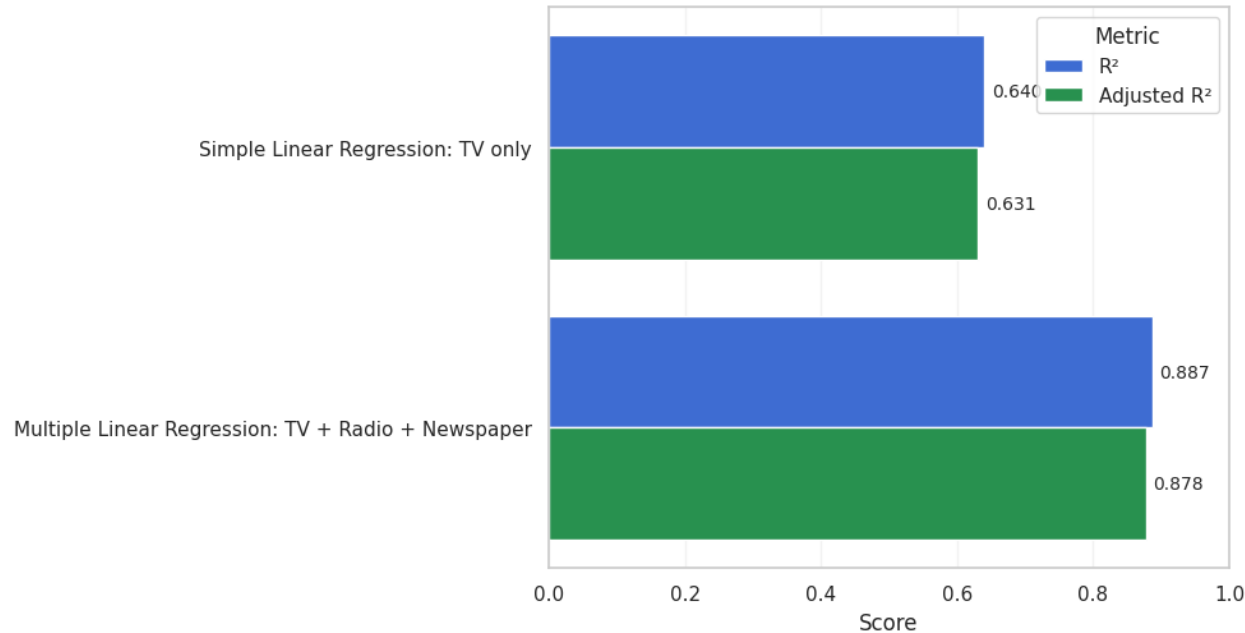
21. BI-Ready Model Comparison Chart This chart turns the comparison table into a print-ready visual. How to read it: higher R^2 and adjusted R^2 mean stronger explained variation. A large gap between simple and multiple models means additional channels add predictive value.

BI interpretation

Use this visual to connect model output with a decision: better fit, stronger driver, more reliable probabilities, a safer threshold, or a clearer diagnostic risk. The chart should support a recommendation rather than remain a technical output.

Does adding Radio and Newspaper improve Sales prediction beyond TV alone?

Model Fit Comparison: Simple vs Multiple Regression



23. BI-Ready Feature Contribution Chart

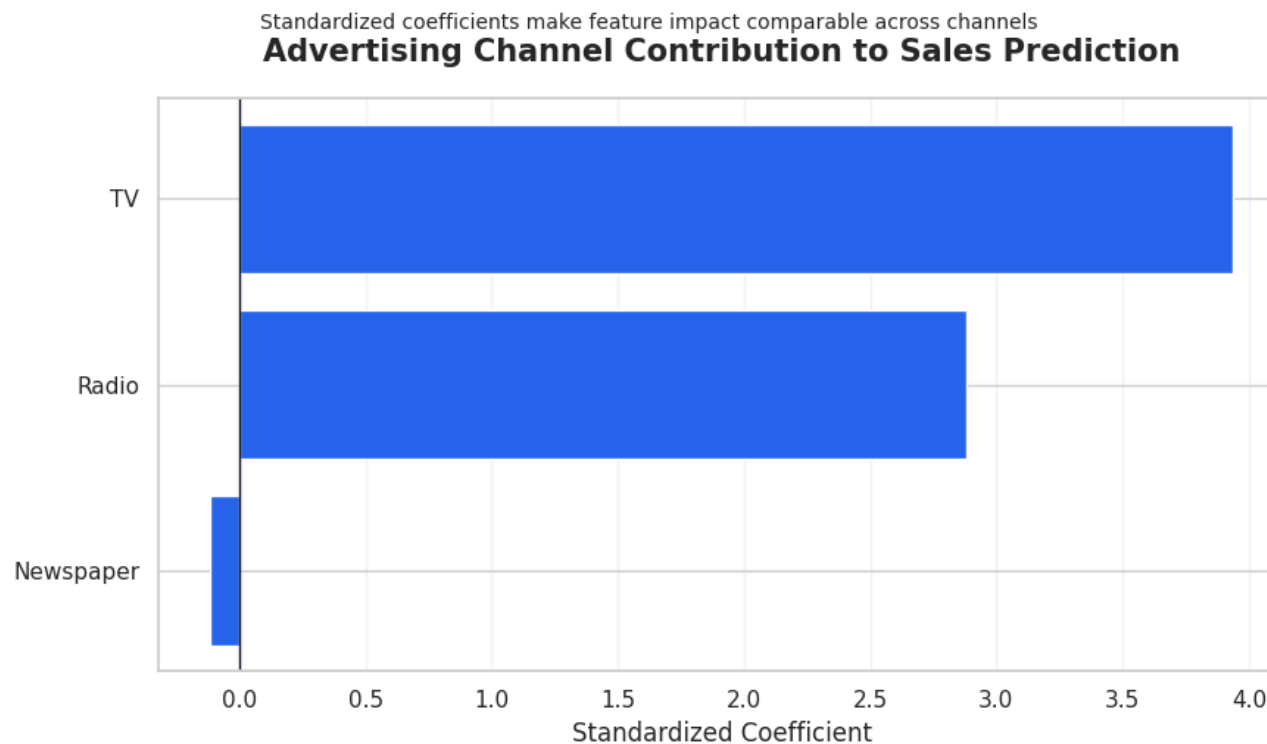
Notebook chart evidence and business interpretation

How to read this chart

23. BI-Ready Feature Contribution Chart This chart shows which advertising channel has the strongest model impact. How to read it: features are sorted by absolute standardized impact. Longer bars mean stronger influence on predicted Sales.

BI interpretation

Use this visual to connect model output with a decision: better fit, stronger driver, more reliable probabilities, a safer threshold, or a clearer diagnostic risk. The chart should support a recommendation rather than remain a technical output.



25. Residual Plot The residual plot checks whether prediction errors

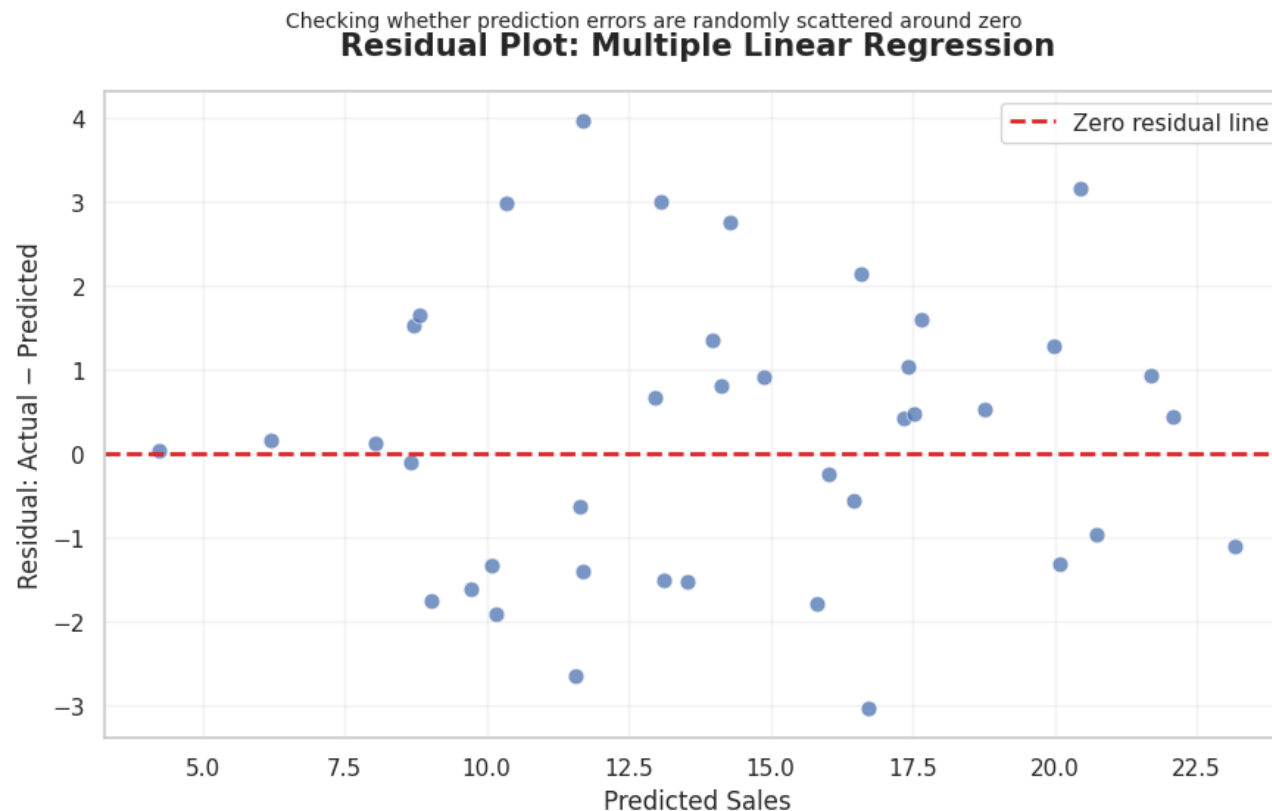
Notebook chart evidence and business interpretation

How to read this chart

25. Residual Plot The residual plot checks whether prediction errors show a pattern. Good sign: residuals are randomly scattered around zero. Warning sign: curved pattern, funnel shape, or systematic trend. This helps check linearity and constant error spread.

BI interpretation

Use this visual to connect model output with a decision: better fit, stronger driver, more reliable probabilities, a safer threshold, or a clearer diagnostic risk. The chart should support a recommendation rather than remain a technical output.

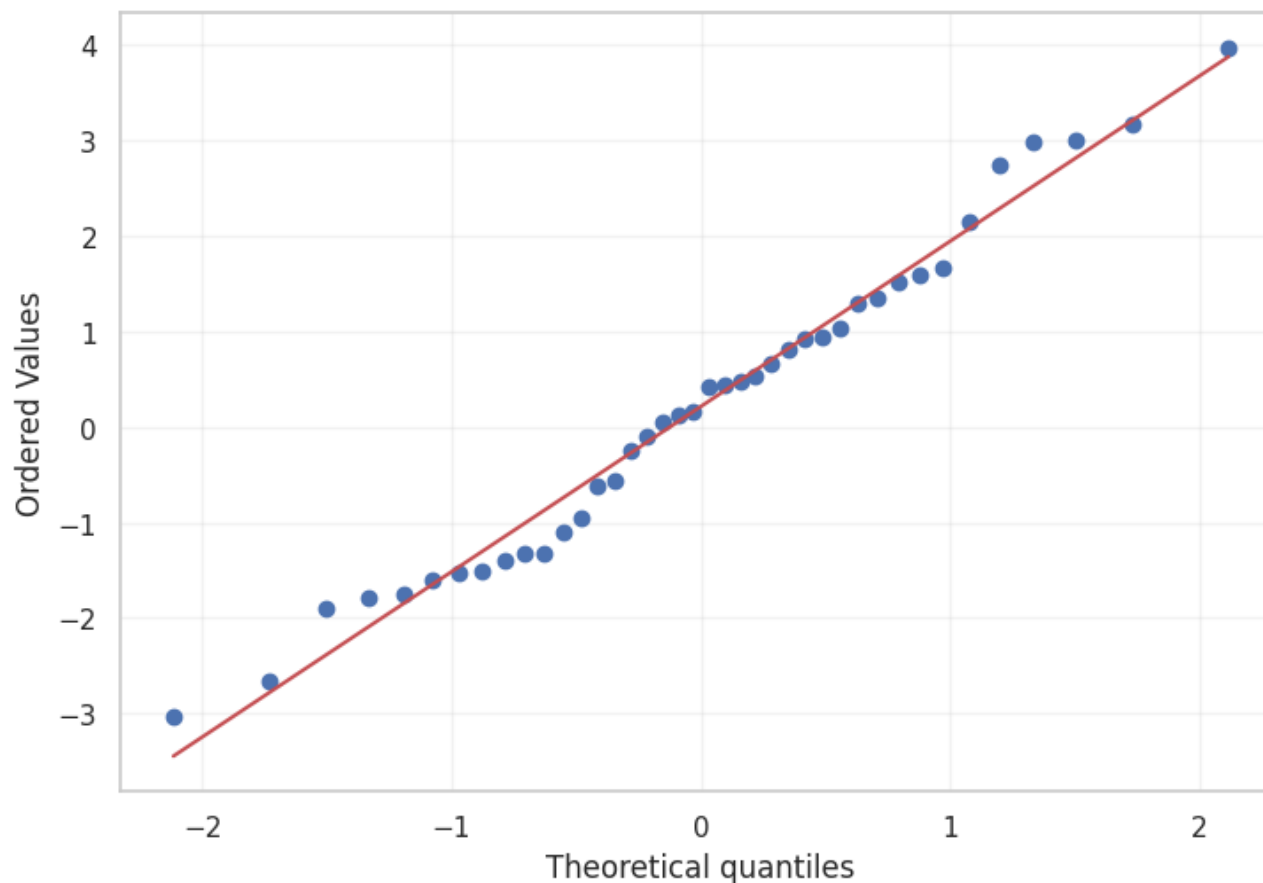


26. QQ Plot for Residual Normality The QQ plot checks whether residuals are

Notebook chart evidence and business interpretation

If points follow the diagonal line, residuals are close to normally distributed

QQ Plot: Residual Normality Check



How to read this chart

26. QQ Plot for Residual Normality The QQ plot checks whether residuals are approximately normally distributed. Simple explanation: if the points follow the diagonal line, residuals are close to normal. If the tails bend far away, the residual distribution may be skewed or heavy-tailed. Why it matters: normality is especially important for statistical inference such as p-values and confidence intervals.

BI interpretation

Use this visual to connect model output with a decision: better fit, stronger driver, more reliable probabilities, a safer threshold, or a clearer diagnostic risk. The chart should support a recommendation rather than remain a technical output.

15. MVD Chart — RMSE Against Alpha The MVD requires a plot of

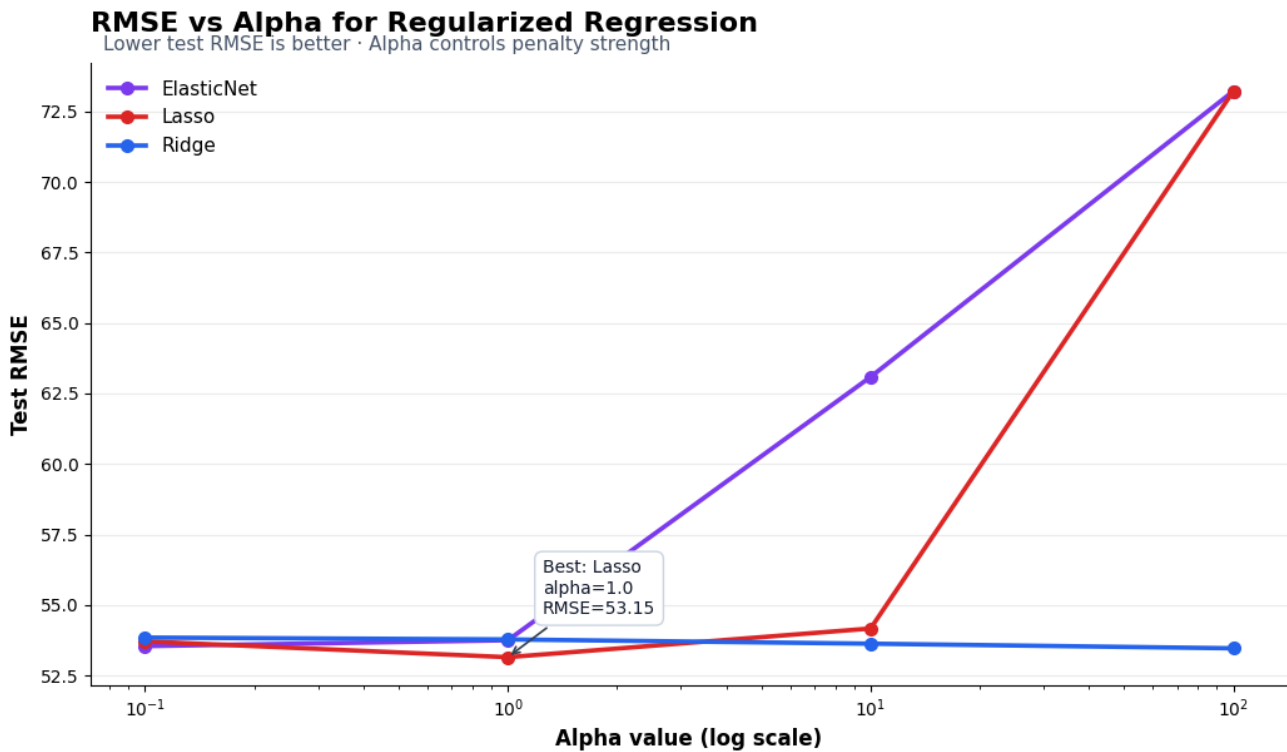
Notebook chart evidence and business interpretation

How to read this chart

15. MVD Chart — RMSE Against Alpha The MVD requires a plot of RMSE against alpha for Ridge, Lasso and ElasticNet. How to read it: lower RMSE is better. If RMSE increases sharply as alpha grows, the model is becoming too restricted and may underfit. Variable flow: results from the Ridge, Lasso and ElasticNet loops are combined into regularized_results_table and used for the chart.

BI interpretation

Use this visual to connect model output with a decision: better fit, stronger driver, more reliable probabilities, a safer threshold, or a clearer diagnostic risk. The chart should support a recommendation rather than remain a technical output.



17. BI-Ready Coefficient Shrinkage Chart

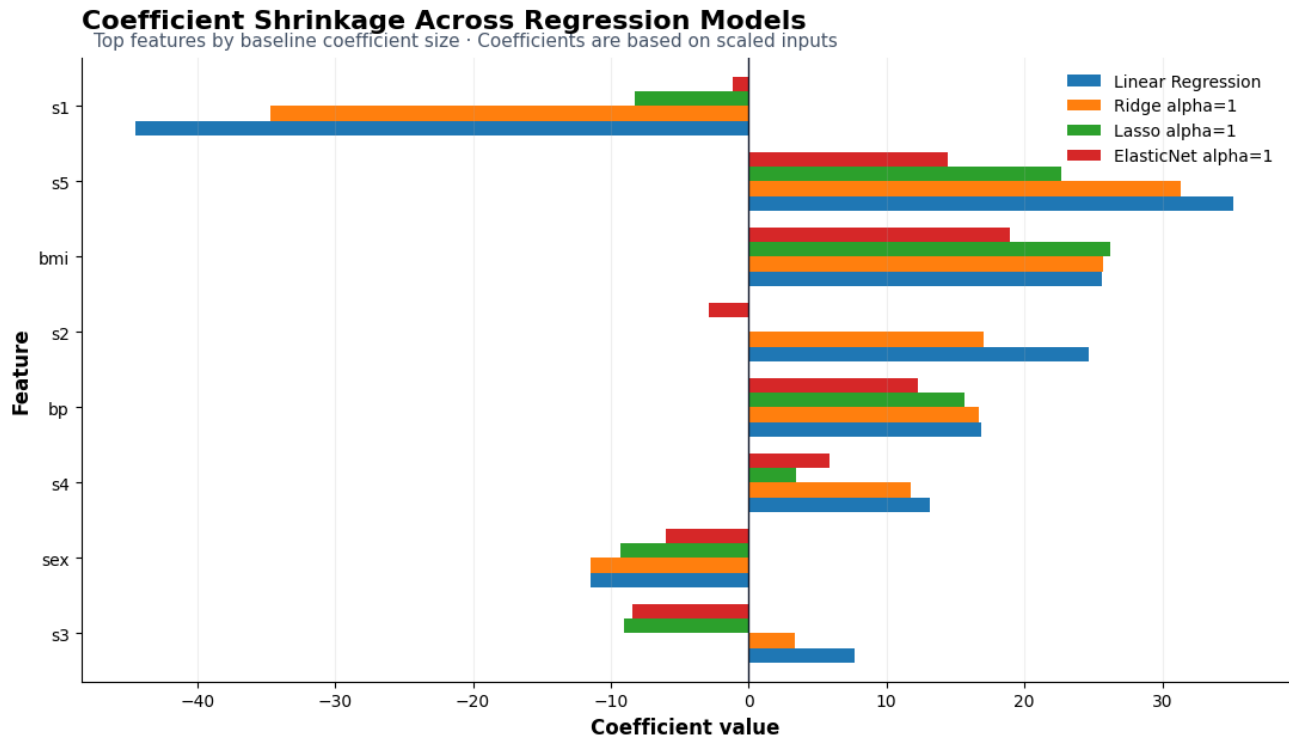
Notebook chart evidence and business interpretation

How to read this chart

17. BI-Ready Coefficient Shrinkage Chart This chart makes coefficient shrinkage visible. It shows the top features by baseline coefficient size and compares how regularization changes them. Business meaning: if a feature remains strong after regularization, it is more stable. If it collapses to zero, it may be weak, redundant or not useful for prediction.

BI interpretation

Use this visual to connect model output with a decision: better fit, stronger driver, more reliable probabilities, a safer threshold, or a clearer diagnostic risk. The chart should support a recommendation rather than remain a technical output.



19. Visualize the Lasso Coefficient Path

Notebook chart evidence and business interpretation

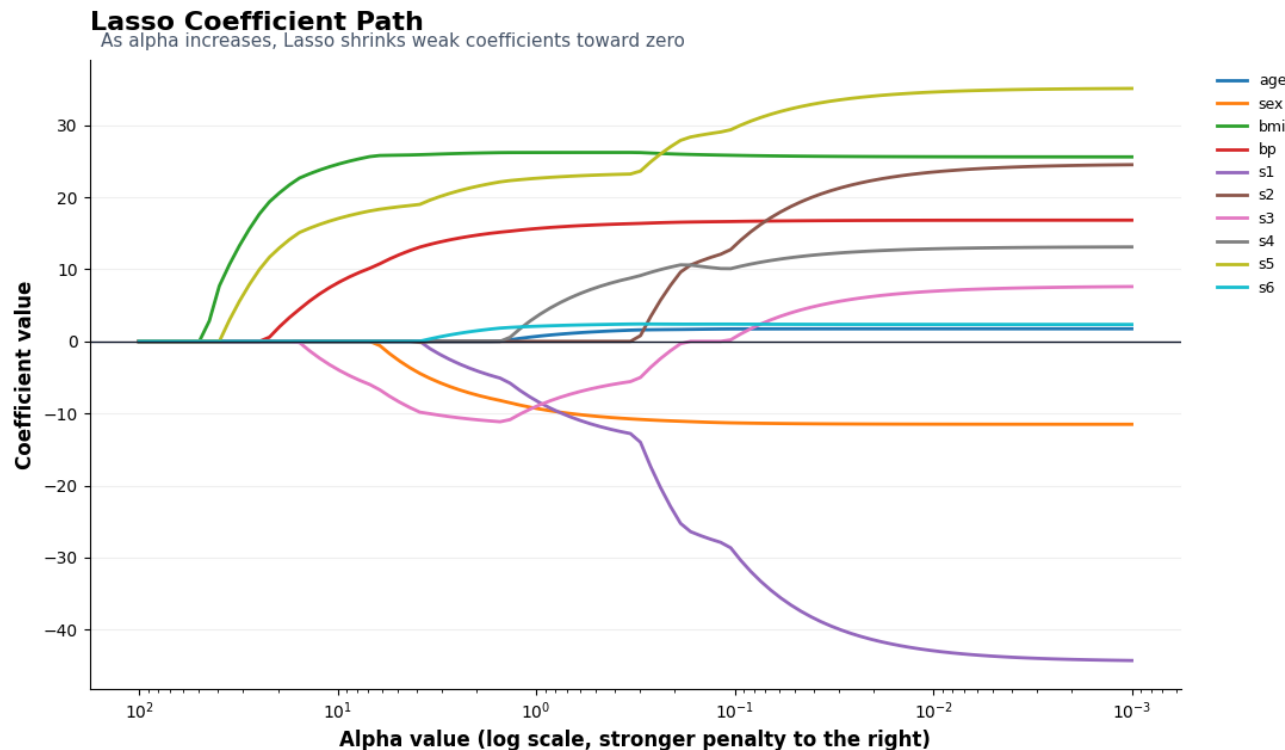
How to read this chart

19. Visualize the Lasso Coefficient Path This chart shows each feature coefficient as alpha changes. The x-axis is alpha on a log scale.

Moving to the right means stronger regularization. How to read it: when a line touches zero and stays there, Lasso has removed that feature from the model for that alpha range.

BI interpretation

Use this visual to connect model output with a decision: better fit, stronger driver, more reliable probabilities, a safer threshold, or a clearer diagnostic risk. The chart should support a recommendation rather than remain a technical output.



23. BI-Ready Final Model Comparison Chart

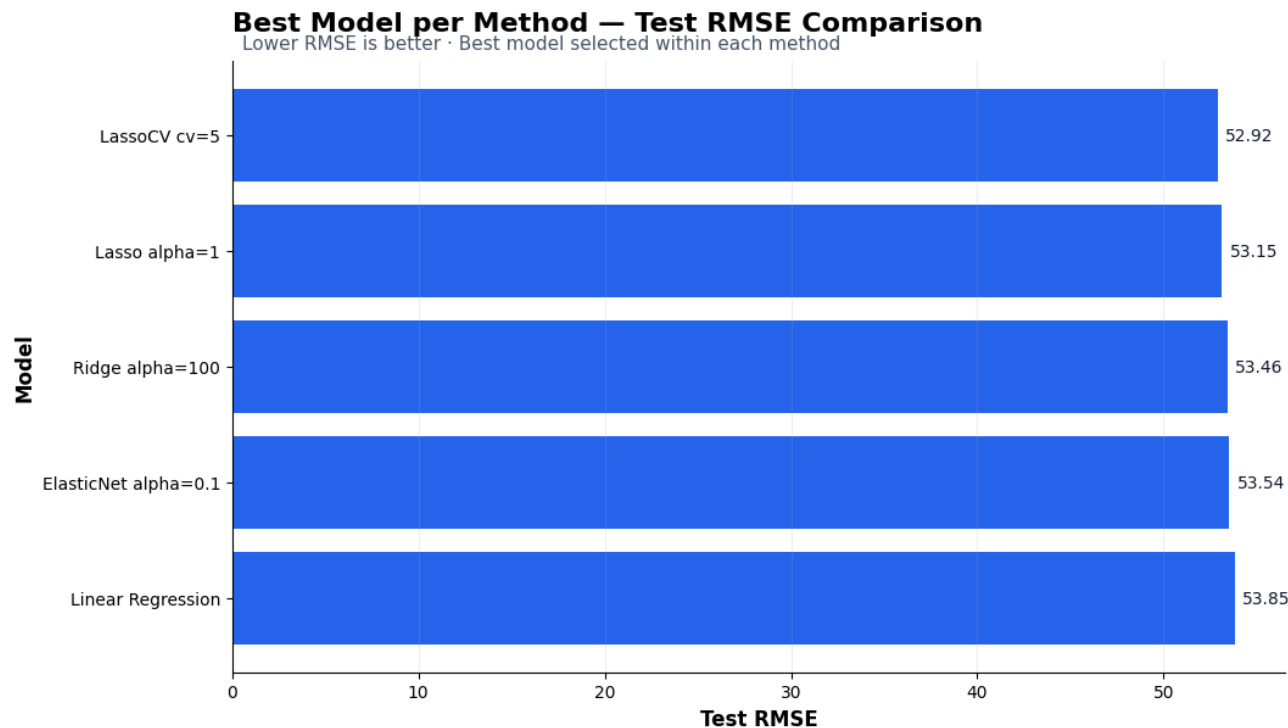
Notebook chart evidence and business interpretation

How to read this chart

23. BI-Ready Final Model Comparison Chart
This chart shows the best version of each method, so the final comparison is easy to read. Business meaning: the recommended model is not always the most complex one. We care about prediction quality, stability and interpretability.

BI interpretation

Use this visual to connect model output with a decision: better fit, stronger driver, more reliable probabilities, a safer threshold, or a clearer diagnostic risk. The chart should support a recommendation rather than remain a technical output.



14. Residual Plot for Linear Regression A residual plot shows whether the model makes systematic errors.

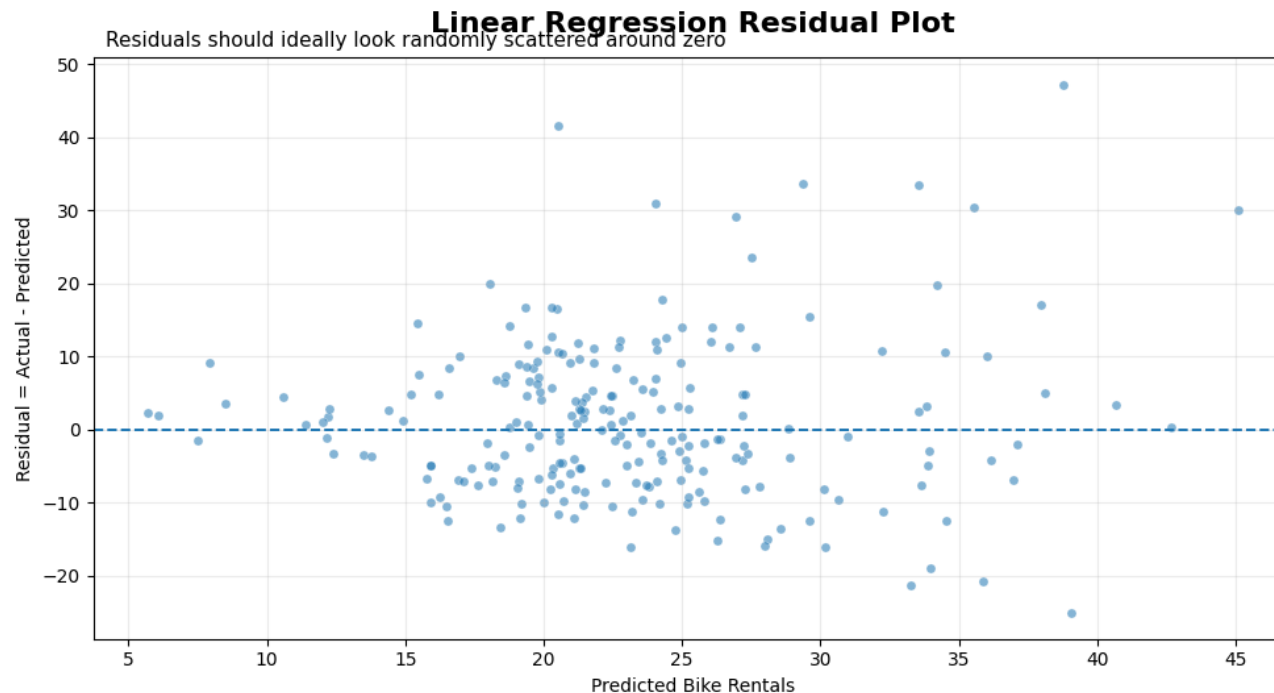
Notebook chart evidence and business interpretation

How to read this chart

14. Residual Plot for Linear Regression A residual plot shows whether the model makes systematic errors. A good residual plot should look like random noise around zero. Patterns suggest that the model is missing structure. Business meaning: if the model underpredicts during high-demand periods, the business may underestimate staffing, inventory or bike availability needs.

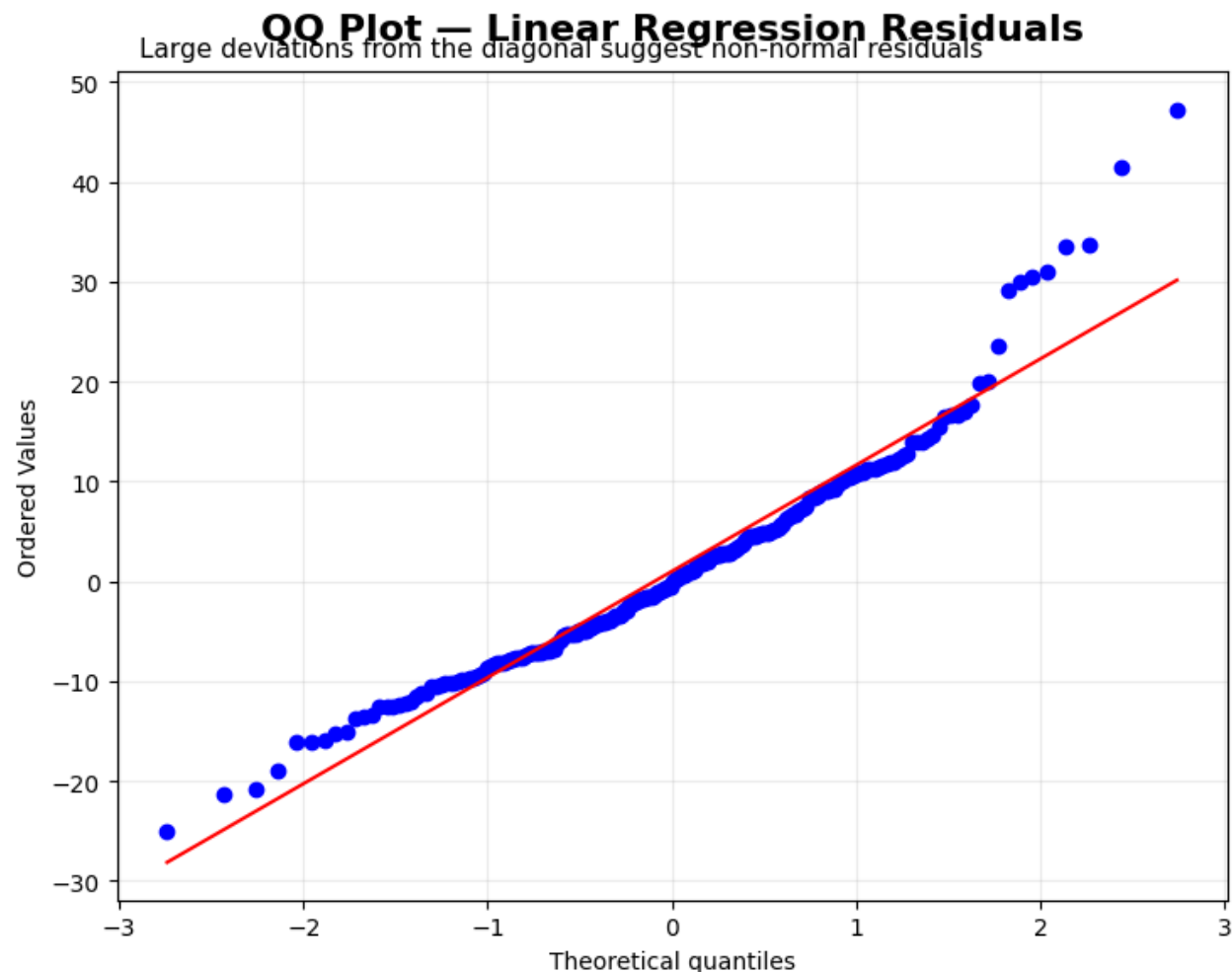
BI interpretation

Use this visual to connect model output with a decision: better fit, stronger driver, more reliable probabilities, a safer threshold, or a clearer diagnostic risk. The chart should support a recommendation rather than remain a technical output.



15. QQ Plot for Residual Normality The MVD specifically asks us to

Notebook chart evidence and business interpretation



How to read this chart

15. QQ Plot for Residual Normality The MVD specifically asks us to check residuals for normality using a QQ plot. A QQ plot compares model residuals with a normal distribution. If the points follow the diagonal line, residuals are roughly normal. Why it matters: count data often produces non-normal residuals. This gives us evidence for trying Poisson and Negative Binomial models.

BI interpretation

Use this visual to connect model output with a decision: better fit, stronger driver, more reliable probabilities, a safer threshold, or a clearer diagnostic risk. The chart should support a recommendation rather than remain a technical output.

22. BI-Ready Model Comparison Chart

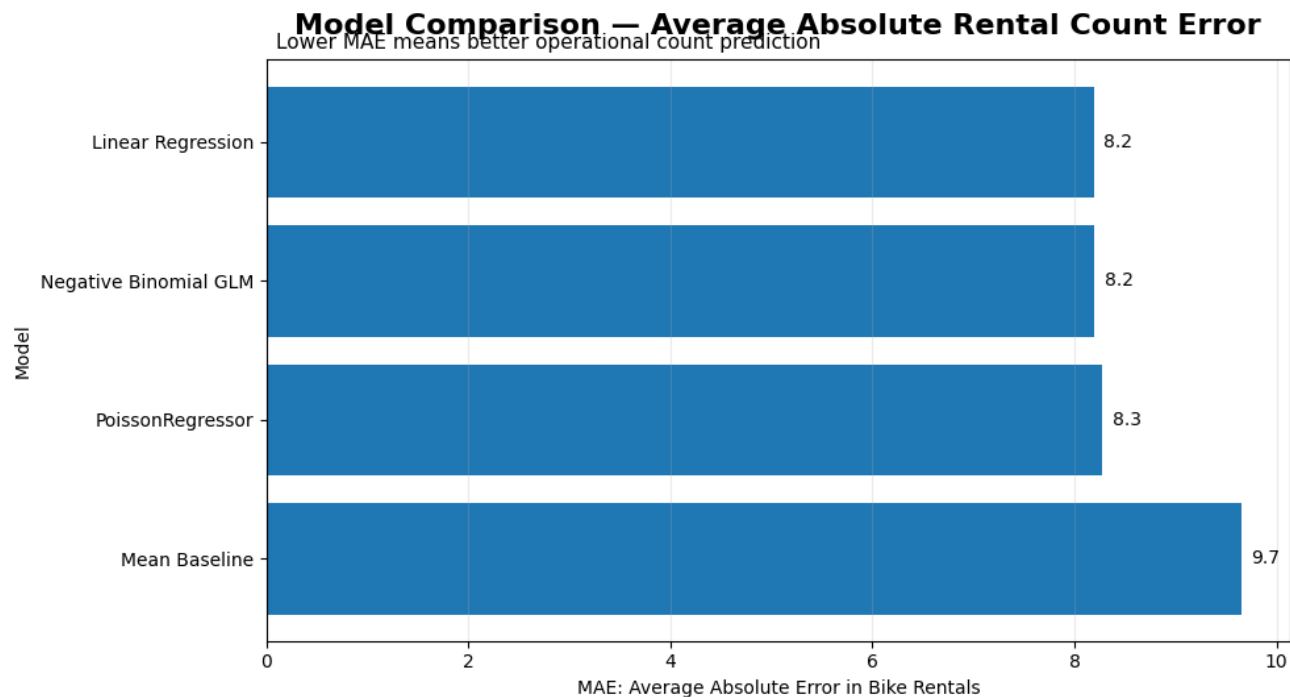
Notebook chart evidence and business interpretation

How to read this chart

22. BI-Ready Model Comparison Chart This chart shows MAE by model. MAE is easy to explain: on average, how many rentals away was the prediction? Business meaning: lower MAE means the model is more useful for planning bike supply, staff scheduling and demand management.

BI interpretation

Use this visual to connect model output with a decision: better fit, stronger driver, more reliable probabilities, a safer threshold, or a clearer diagnostic risk. The chart should support a recommendation rather than remain a technical output.



23. Actual vs Predicted Counts

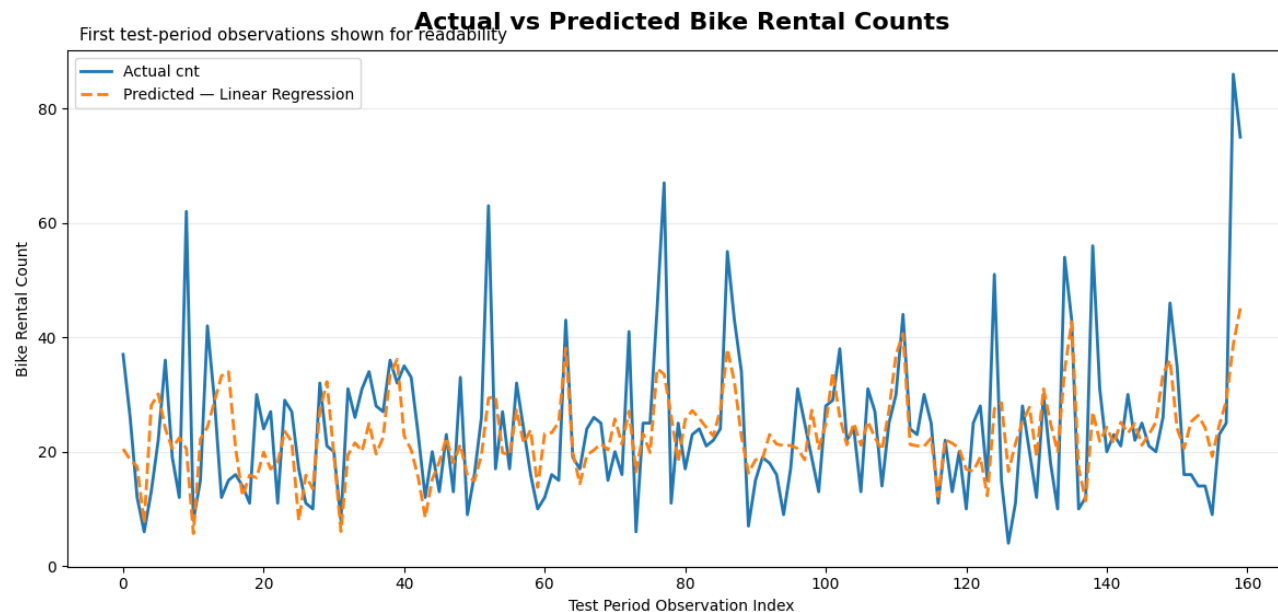
Notebook chart evidence and business interpretation

How to read this chart

23. Actual vs Predicted Counts This chart compares actual counts with predictions from the best MAE model. It helps us see whether the model follows the movement of demand or only predicts an average level. Business meaning: good demand models should not only have low average error; they should also track peaks and dips reasonably well.

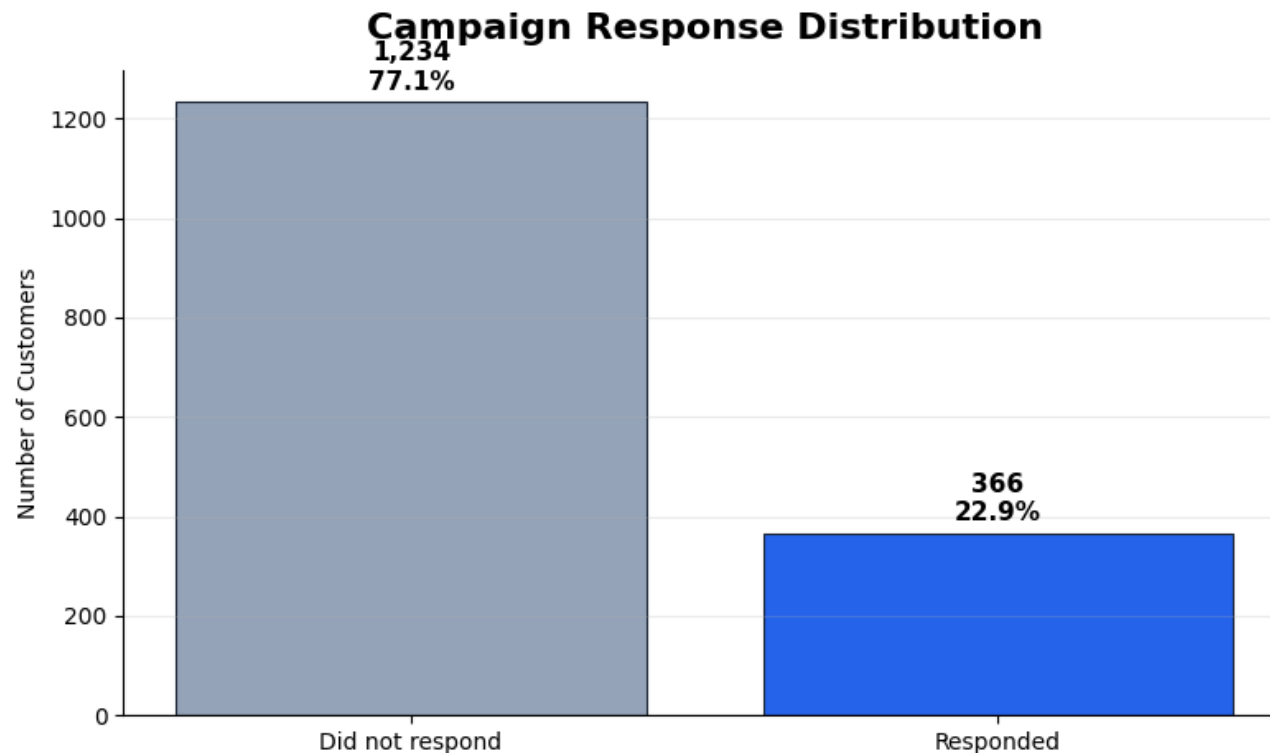
BI interpretation

Use this visual to connect model output with a decision: better fit, stronger driver, more reliable probabilities, a safer threshold, or a clearer diagnostic risk. The chart should support a recommendation rather than remain a technical output.



7. Visualize the Target Distribution The target distribution tells u

Notebook chart evidence and business interpretation



How to read this chart

7. Visualize the Target Distribution The target distribution tells us whether the classes are balanced or imbalanced. This matters because accuracy can be misleading when one class is much more common than the other. Business meaning: if only a small share of customers respond, the model must be judged by precision, recall, PR-AUC, and lift — not only accuracy.

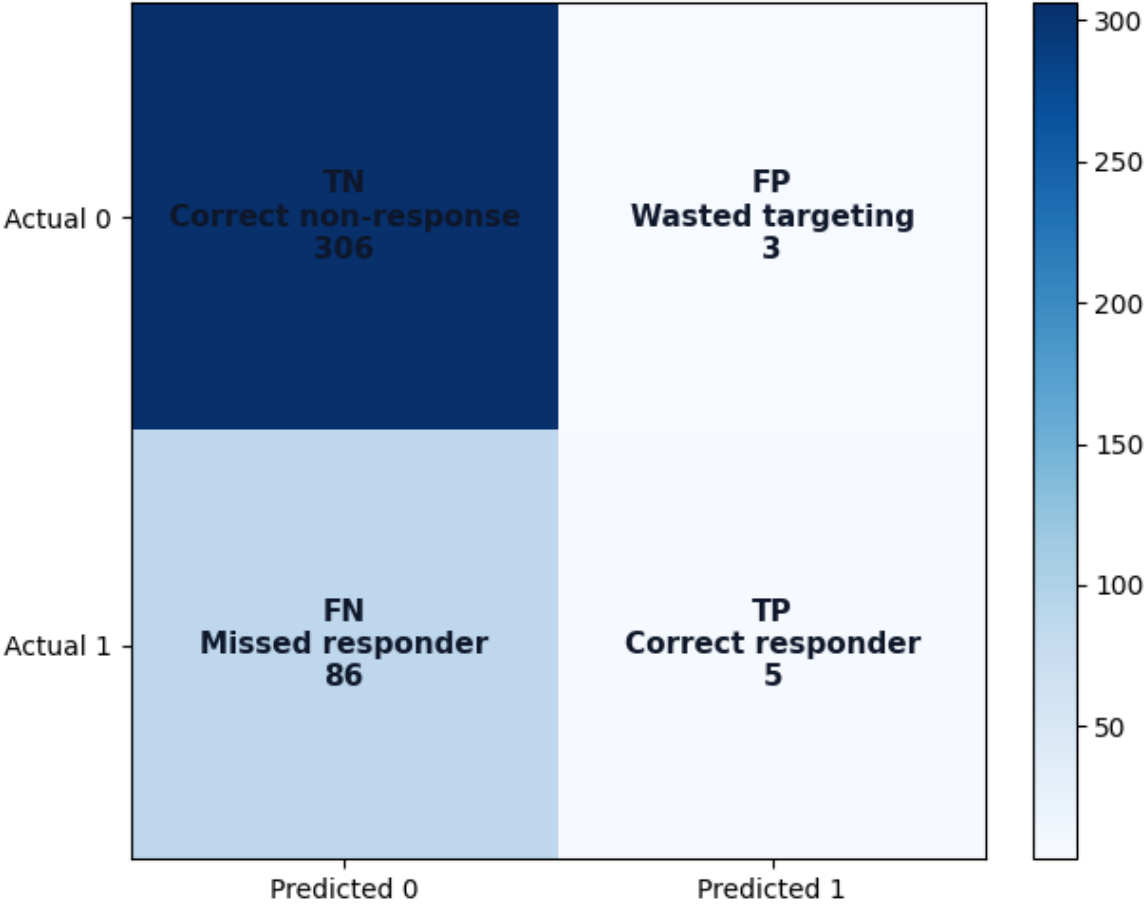
BI interpretation

Use this visual to connect model output with a decision: better fit, stronger driver, more reliable probabilities, a safer threshold, or a clearer diagnostic risk. The chart should support a recommendation rather than remain a technical output.

14. Confusion Matrix The confusion matrix shows where the model

Notebook chart evidence and business interpretation

Confusion Matrix — Baseline Logistic Regression



How to read this chart

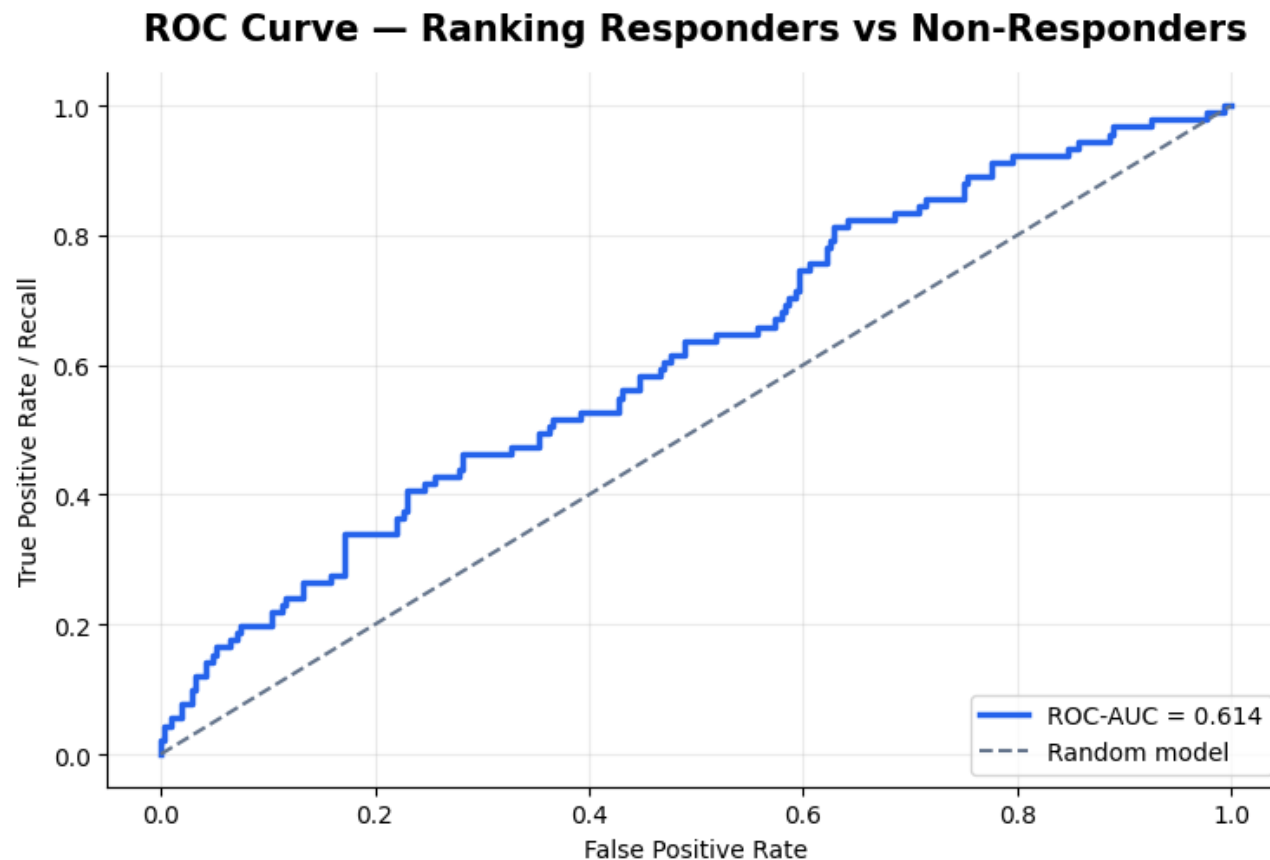
14. Confusion Matrix The confusion matrix shows where the model is right and wrong. It converts abstract metrics into clear business outcomes. Business meaning: false positives are customers we target who do not respond. False negatives are real responders we missed. The best threshold depends on which mistake is more expensive.

BI interpretation

Use this visual to connect model output with a decision: better fit, stronger driver, more reliable probabilities, a safer threshold, or a clearer diagnostic risk. The chart should support a recommendation rather than remain a technical output.

15. ROC Curve The ROC curve shows how well the model ranks re

Notebook chart evidence and business interpretation



How to read this chart

15. ROC Curve The ROC curve shows how well the model ranks responders above non-responders across many thresholds.

Interpretation: ROC-AUC close to 1 is strong. ROC-AUC close to 0.5 means the model is not much better than random ranking.

BI interpretation

Use this visual to connect model output with a decision: better fit, stronger driver, more reliable probabilities, a safer threshold, or a clearer diagnostic risk. The chart should support a recommendation rather than remain a technical output.

16. Precision-Recall Curve The Precision-Recall curve is often more

Notebook chart evidence and business interpretation

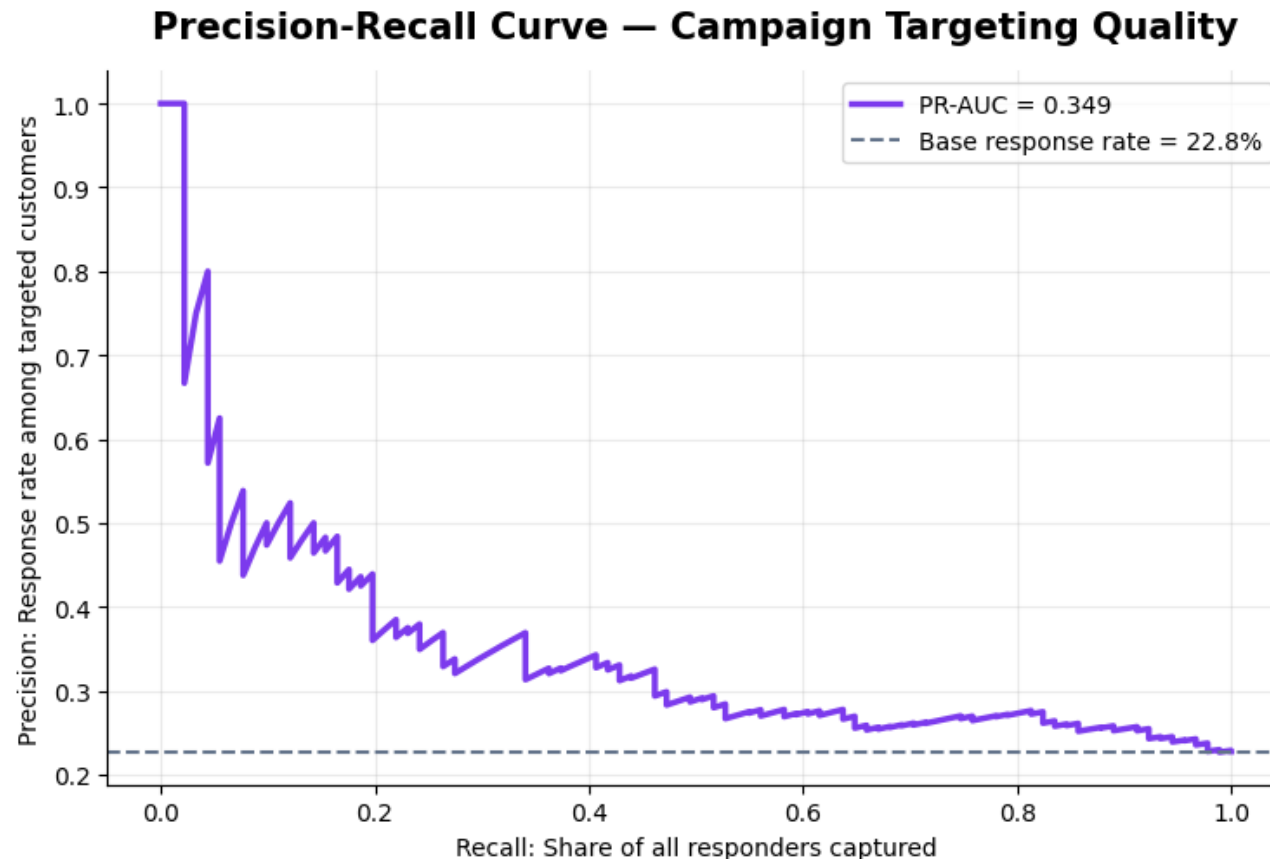
How to read this chart

16. Precision-Recall Curve The Precision-Recall curve is often more useful than ROC when the positive class is less common.

Business meaning: it shows the trade-off between targeting cleaner leads and finding more potential responders.

BI interpretation

Use this visual to connect model output with a decision: better fit, stronger driver, more reliable probabilities, a safer threshold, or a clearer diagnostic risk. The chart should support a recommendation rather than remain a technical output.



17. Visualize the Sigmoid Function The sigmoid function is the heart of Logistic Regression.

Notebook chart evidence and business interpretation

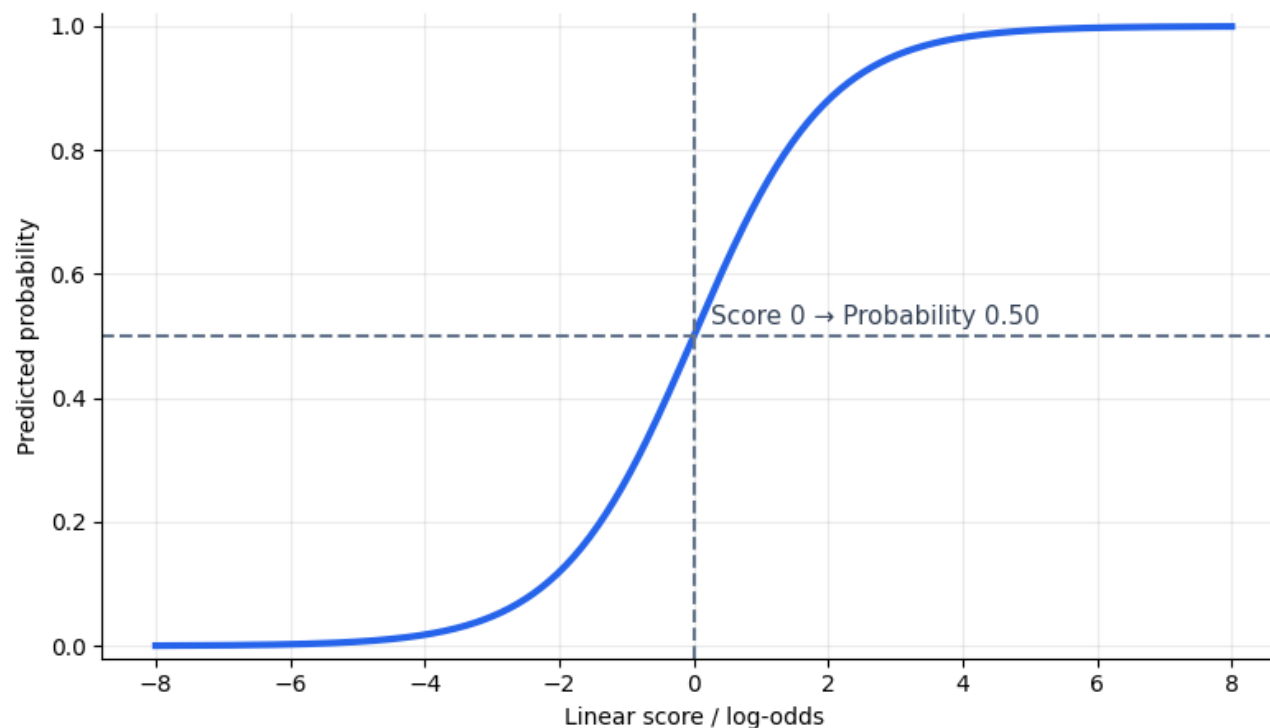
How to read this chart

17. Visualize the Sigmoid Function The sigmoid function is the heart of Logistic Regression. It converts any linear score into a probability between 0 and 1. Simple meaning: very negative scores become probabilities near 0, very positive scores become probabilities near 1, and score 0 becomes probability 0.50.

BI interpretation

Use this visual to connect model output with a decision: better fit, stronger driver, more reliable probabilities, a safer threshold, or a clearer diagnostic risk. The chart should support a recommendation rather than remain a technical output.

Sigmoid Function — Converts Score into Probability



19. Coefficient Impact Chart

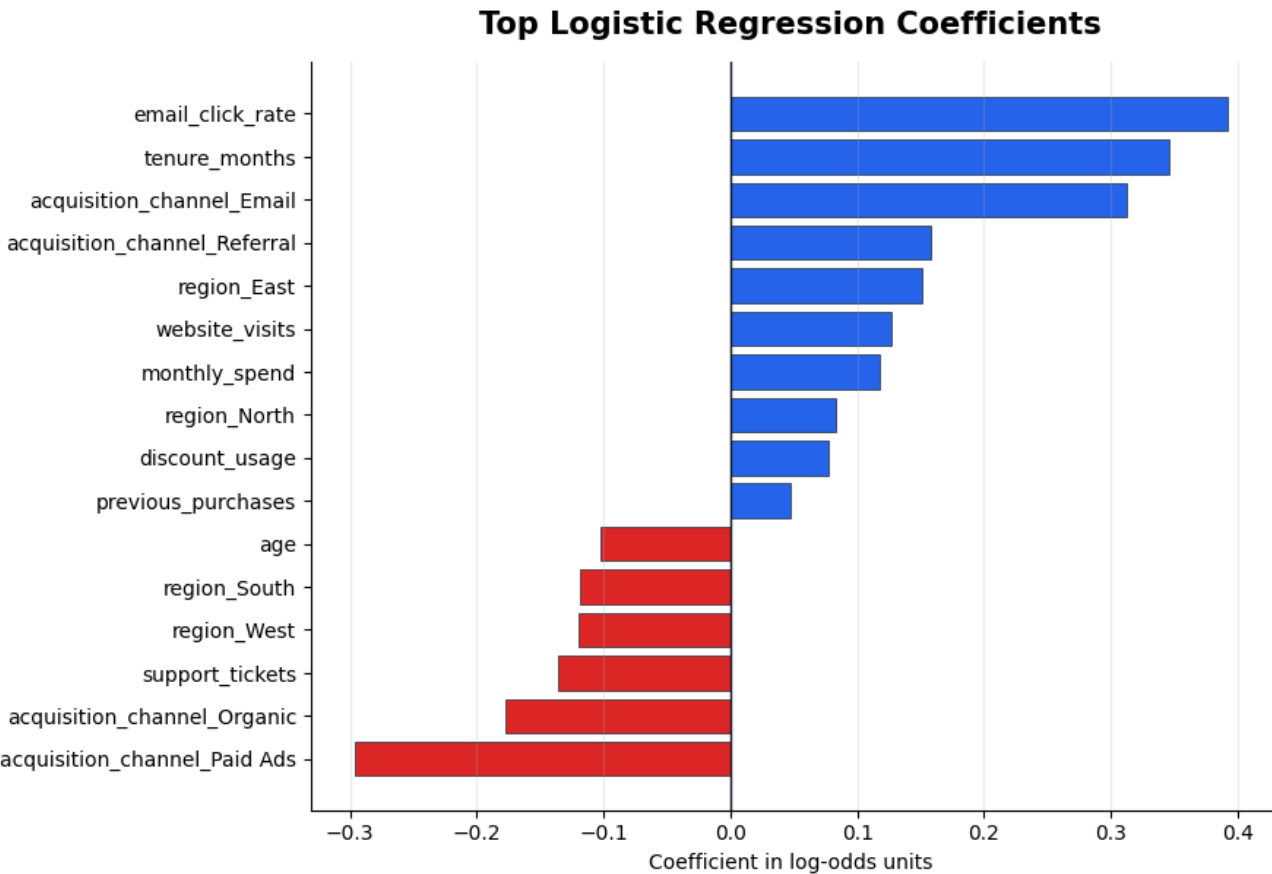
Notebook chart evidence and business interpretation

How to read this chart

19. Coefficient Impact Chart This chart shows the strongest positive and negative coefficient effects. It helps explain which customer signals push the model toward response or non-response. Important caution: coefficients are affected by scaling and encoding. They are useful for interpretation, but they are not the same as causal proof.

BI interpretation

Use this visual to connect model output with a decision: better fit, stronger driver, more reliable probabilities, a safer threshold, or a clearer diagnostic risk. The chart should support a recommendation rather than remain a technical output.



21. Threshold Trade-Off Chart

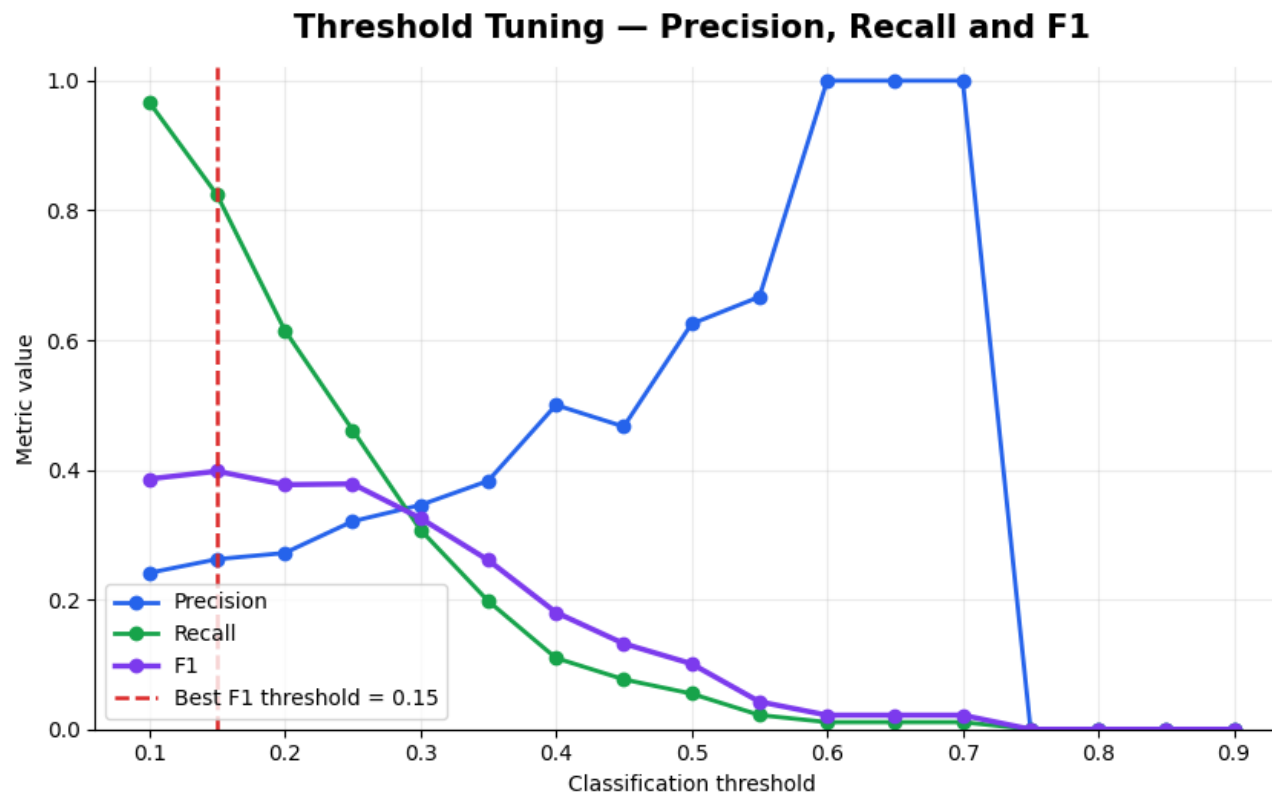
Notebook chart evidence and business interpretation

How to read this chart

21. Threshold Trade-Off Chart This chart shows how precision, recall and F1 change as the threshold changes. It makes the business trade-off visible. Business interpretation: if campaign cost is high, prefer higher precision. If missing responders is costly, prefer higher recall.

BI interpretation

Use this visual to connect model output with a decision: better fit, stronger driver, more reliable probabilities, a safer threshold, or a clearer diagnostic risk. The chart should support a recommendation rather than remain a technical output.



25. Check Probability Calibration Calibration checks whether predicted

Notebook chart evidence and business interpretation

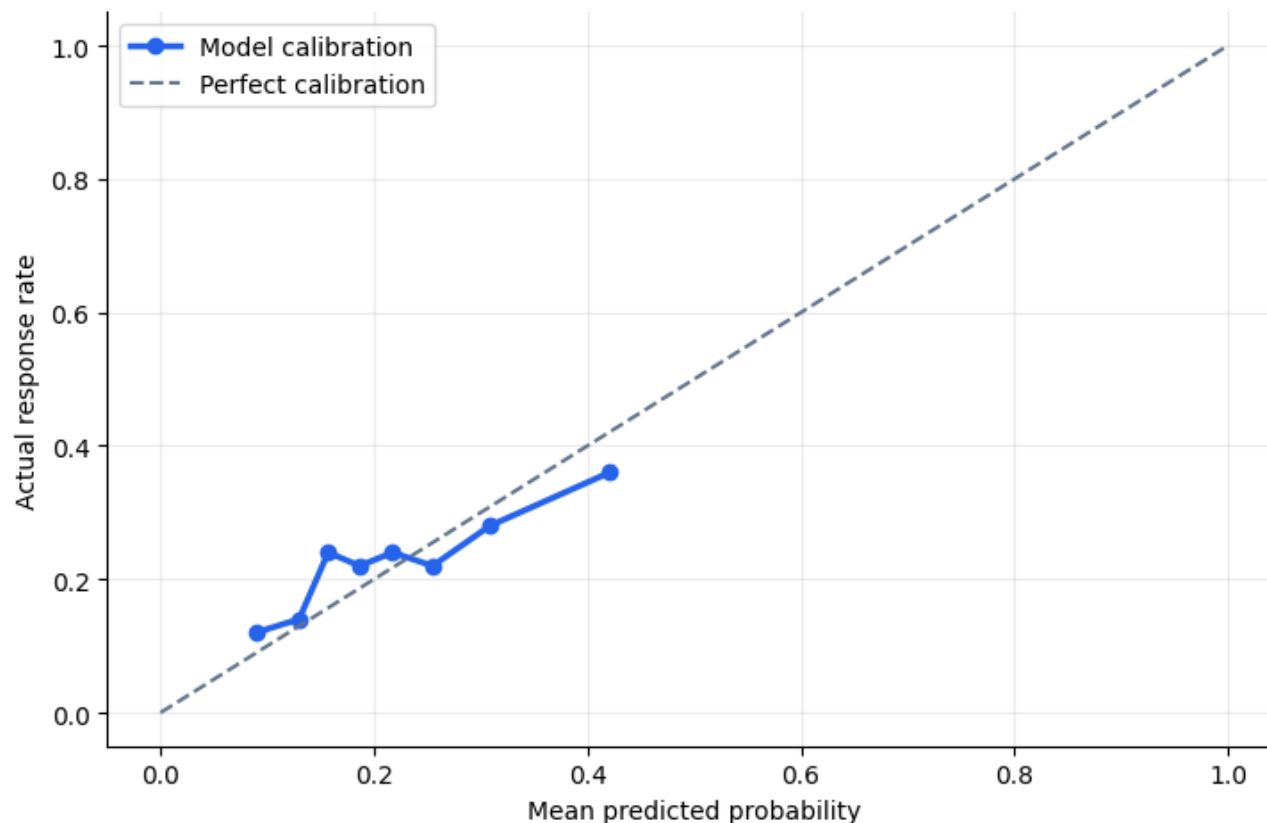
How to read this chart

25. Check Probability Calibration Calibration checks whether predicted probabilities are trustworthy. For example, among customers scored around 70%, roughly 70% should respond if the model is well calibrated. Business meaning: calibrated probabilities are important if the model score is used for budgeting, forecasting campaign ROI, or ranking customers in a dashboard.

BI interpretation

Use this visual to connect model output with a decision: better fit, stronger driver, more reliable probabilities, a safer threshold, or a clearer diagnostic risk. The chart should support a recommendation rather than remain a technical output.

Calibration Curve — Are Probabilities Trustworthy?



27. Lift Chart The lift chart makes the business value visible. It shows which scored groups are worth targeting first. Business interpretation: if the top decile has much higher lift than average, the model can reduce campaign waste by prioritizing high-probability customers.

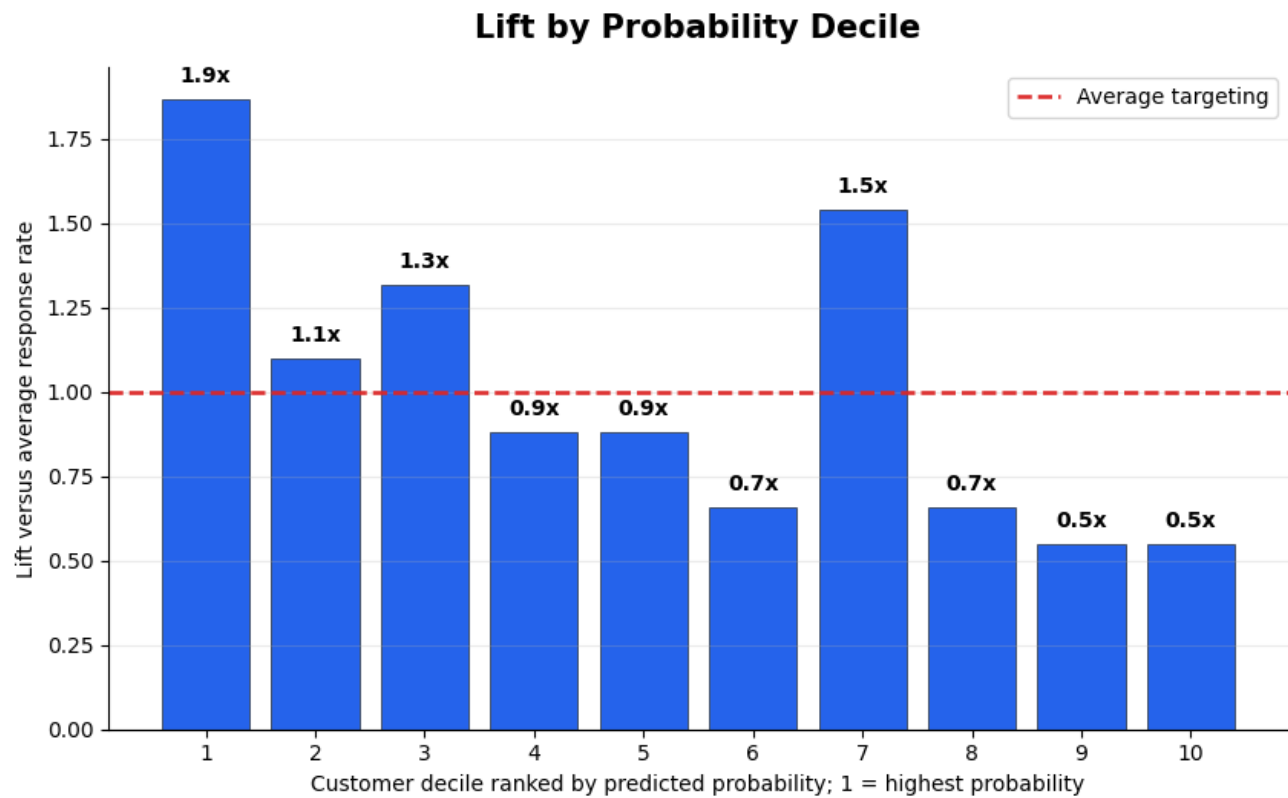
Notebook chart evidence and business interpretation

How to read this chart

27. Lift Chart The lift chart makes the business value visible. It shows which scored groups are worth targeting first. Business interpretation: if the top decile has much higher lift than average, the model can reduce campaign waste by prioritizing high-probability customers.

BI interpretation

Use this visual to connect model output with a decision: better fit, stronger driver, more reliable probabilities, a safer threshold, or a clearer diagnostic risk. The chart should support a recommendation rather than remain a technical output.



Code cell 8 — Run gradient descent and plot loss

Notebook chart evidence and business interpretation

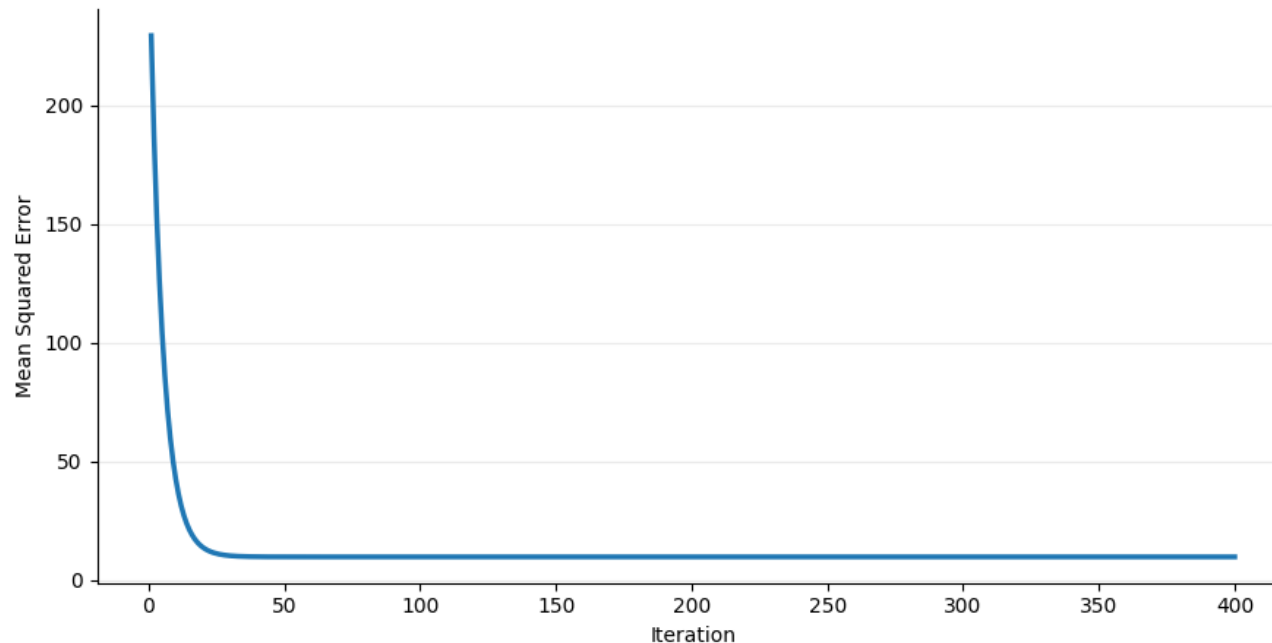
How to read this chart

Code cell 8 — Run gradient descent and plot loss This cell performs gradient descent manually. The loss should decrease over iterations, showing how the model learns coefficients by repeatedly reducing squared error.

BI interpretation

Use this visual to connect model output with a decision: better fit, stronger driver, more reliable probabilities, a safer threshold, or a clearer diagnostic risk. The chart should support a recommendation rather than remain a technical output.

Gradient Descent Learning Curve



Code cell 9 — Interpolation vs extrapolation Interpolation means

Notebook chart evidence and business interpretation

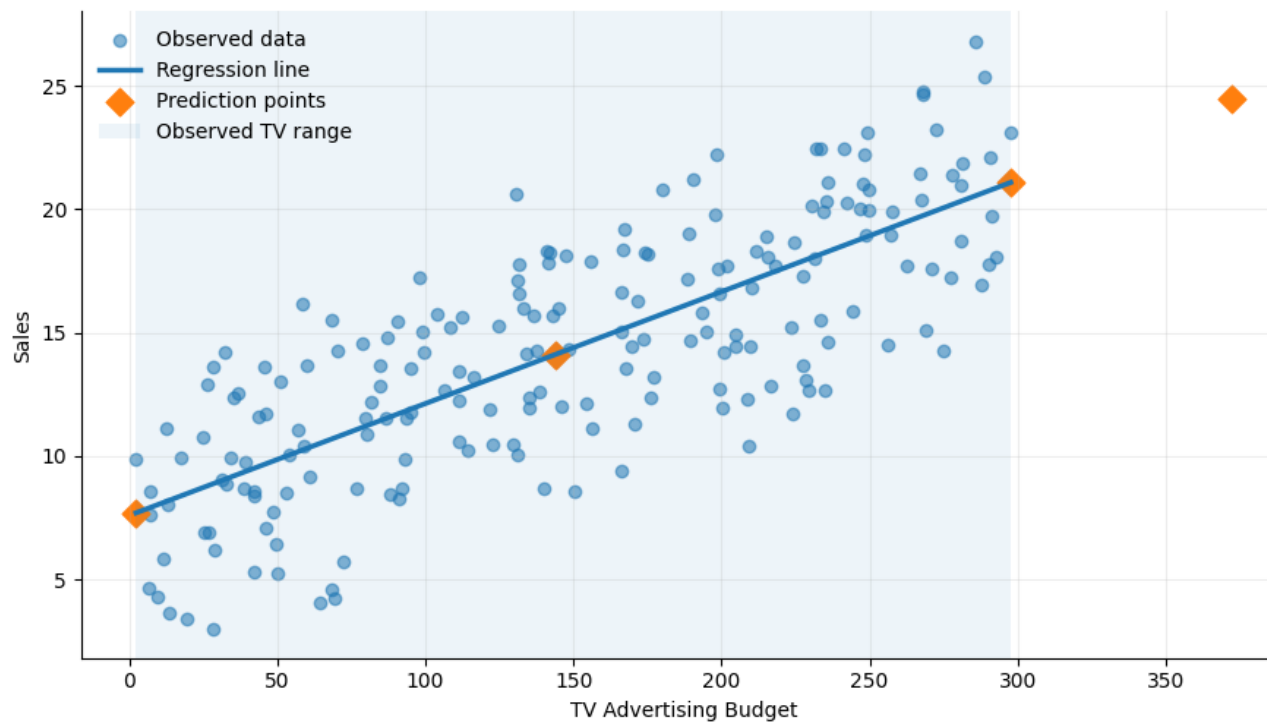
How to read this chart

Code cell 9 — Interpolation vs extrapolation
Interpolation means predicting inside the observed data range. Extrapolation means predicting outside the observed data range. Regression can technically do both, but extrapolation is riskier because the model has not seen that region of data.

BI interpretation

Use this visual to connect model output with a decision: better fit, stronger driver, more reliable probabilities, a safer threshold, or a clearer diagnostic risk. The chart should support a recommendation rather than remain a technical output.

Interpolation vs Extrapolation



Code cell 12 — Homoscedasticity vs heteroscedasticity visual check

Notebook chart evidence and business interpretation

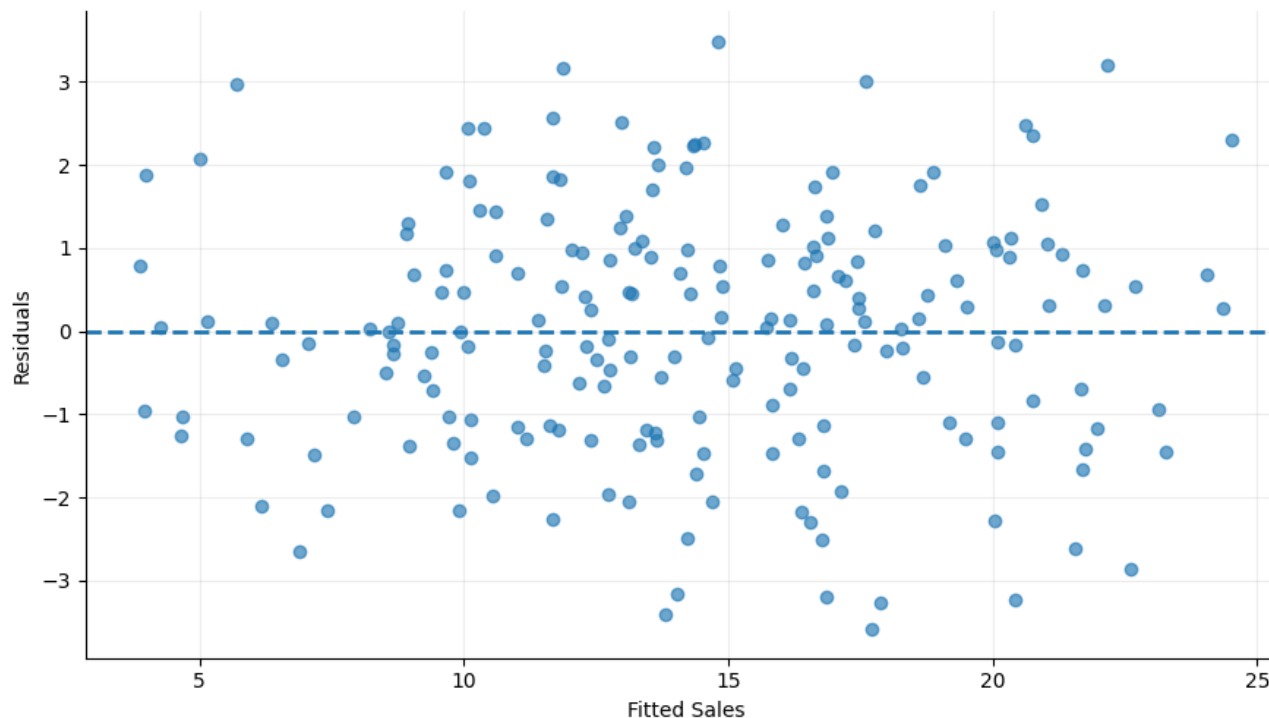
How to read this chart

Code cell 12 — Homoscedasticity vs heteroscedasticity visual check
Homoscedasticity means residuals have roughly equal spread across prediction levels.
Heteroscedasticity means the spread changes, often forming a funnel shape. This matters because unequal spread can make standard errors and p-values less trustworthy.

BI interpretation

Use this visual to connect model output with a decision: better fit, stronger driver, more reliable probabilities, a safer threshold, or a clearer diagnostic risk. The chart should support a recommendation rather than remain a technical output.

Residual Plot: Checking Equal Error Spread



Code cell 17 — Cook's distance and leverage chart

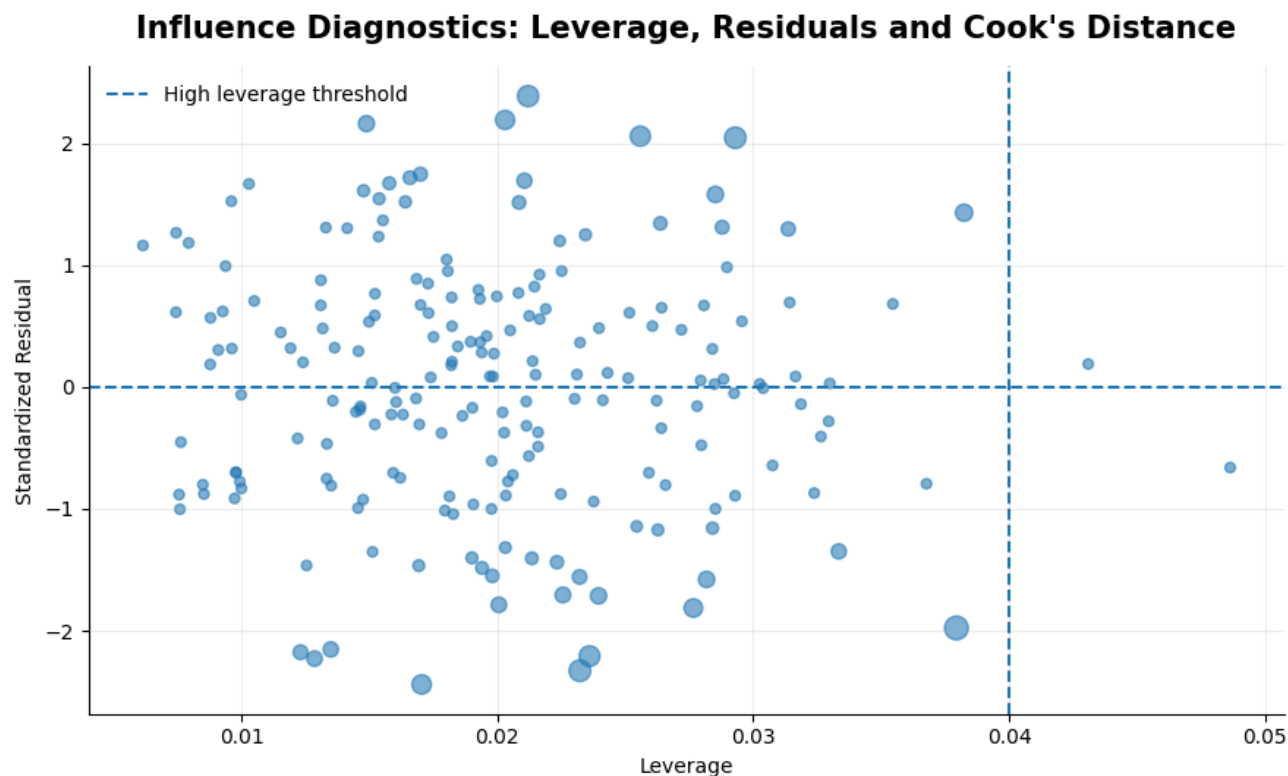
Notebook chart evidence and business interpretation

How to read this chart

Code cell 17 — Cook's distance and leverage chart This chart shows which rows deserve attention. Rows with high leverage and large residuals are often the most important to review.

BI interpretation

Use this visual to connect model output with a decision: better fit, stronger driver, more reliable probabilities, a safer threshold, or a clearer diagnostic risk. The chart should support a recommendation rather than remain a technical output.



Final BI recommendation

Use the model that matches the decision and target type

Sales driver explanation

Multiple Linear Regression with statsmodels OLS

Stable numeric prediction with correlated features

Ridge or ElasticNet

Feature selection and simpler numeric model

LassoCV

Count target with overdispersion

Negative Binomial GLM

Campaign targeting and customer response scoring

Logistic Regression with threshold and lift analysis